

Technical Document #547: Retrieval Augmented Generation - Comprehensive Overview

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Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Hallucination detection identifies when LLMs generate factually incorrect information. It is critical for maintaining trust in RAG systems.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Robotics combines mechanical engineering, electrical engineering, and computer science. Modern robots can perform complex tasks in unstructured environments.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Vector databases store dense embeddings for semantic search.
- Re-ranking models score retrieved documents for relevance to the query.